

Web–Field Model of Cognitive Trait Distributions and Social Evolution (Working Theory)

Note: the core theory of CMTR (cognitive manifold trait ratio) is being used to create non-pathologizing pedagogical evaluations and tools to accelerate student learning and student-teacher/teacher-preset and student-preset matching for Plurimo; separately I plan to continue applying RICD to the overall theory of trait distributions and social evolution, which I'm dubbing (as a general term) "systems cognition."

1. Core Ontology: Trait Manifold

Human cognition and behavior are modeled as a **continuous, high-dimensional trait manifold**.

- Individuals are not discrete "types," but **points or trajectories** within this manifold.
- The manifold contains multiple overlapping regions of higher or lower probability density.
- Traits associated with constructs like ADHD, autism, giftedness, or disability are not separate categories, but **partially overlapping regions of this continuous space**.

Key principle:

Cognition is fundamentally continuous, not categorical.

2. Dynamic Distribution (Flux + Homeostasis)

The population distribution over the manifold is **not static**.

Two competing dynamics govern it:

A. Flux (change)

- Genetic variation, development, environment, and culture continuously shift:
 - individual trajectories
 - regional density in the manifold
- These changes occur across multiple timescales (developmental, generational, historical).

B. Homeostasis (balance)

- Stability emerges not from fixed categories, but from the **relative balance (ratios) of population densities across regions of the manifold**.
- The system tends to maintain functional coherence through these ratios rather than absolute stability of any single region.

Homeostasis = stable ratios of distributed cognitive variation, not fixed trait categories.

3. Multi-Manifold Structure (Web Component)

Instead of a single homogeneous distribution, the system is modeled as a **web of interacting manifolds**:

- Each “manifold region” represents a functional clustering of traits (not a discrete type).
- These regions overlap and interact.
- Individuals may occupy intermediate or shifting positions across regions.

This creates a **networked structure of partially overlapping cognitive distributions**, rather than isolated categories.

4. Social and Cultural Coupling

Social values as dynamic feedback variables

Social systems interact bidirectionally with the manifold:

- Social values influence:
 - which regions of the manifold are rewarded or suppressed
 - developmental environments
 - expression of traits in behavior and outcomes
- The distribution of traits influences:
 - group psychology
 - institutional design

- cultural values over time

Social values and trait distributions form a feedback loop.

5. Group Psychology as the Mediating Layer

Between individual traits and institutions lies **group psychology**, which mediates system-level effects:

- conformity pressure
- status hierarchies
- norm formation
- rigidity vs flexibility in collective behavior

This layer determines how cognitive diversity is translated into:

- coordination efficiency
 - institutional structure
 - cultural stability or fragility
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6. System Stability and Collapse Dynamics

System behavior depends on the interaction of manifold distributions and group psychology:

Balanced regime

- diverse cognitive regions are functionally integrated
- system remains adaptive and flexible

Imbalanced regime

- certain regions become over- or under-represented or structurally constrained
- group psychology shifts toward rigidity and hierarchical control
- systems prioritize stability of form over adaptive function

Outcome dynamics

- rigidity increases fragility
- systems may collapse or transition into a new stable configuration

Collapse is interpreted as a phase transition to a new equilibrium distribution.

7. Phase Transitions and Paradigm Shifts

When imbalance reaches critical thresholds:

- systems may reorganize into a new stable configuration
 - this represents a **shift in the optimal ratio structure across manifolds**
 - “new paradigms” correspond to new equilibrium distributions rather than linear change
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8. Exceptional Individuals and Tail Dynamics

Within the manifold:

- rare or extreme trajectories can have disproportionate systemic impact
- these individuals can:
 - introduce structural innovations
 - reshape cultural or institutional trajectories
 - amplify or destabilize existing equilibria

They are not separate mechanisms, but **low-probability, high-impact regions of the same space**.

9. Intervention Logic (Implied Framework)

Direct control of “types” is not assumed.

Instead, change occurs through:

- altering incentives and institutional structures
- reshaping developmental environments
- modifying social reinforcement patterns

These interventions:

- change which regions of the manifold are expressed or reinforced
- shift density distributions over time
- indirectly affect system stability and evolution

Intervention operates through environmental and cultural reshaping, not categorical manipulation.

10. Web–Field Architecture (Meta-Model)

The full system is best understood as:

- a **continuous cognitive trait field (manifold)**
- embedded within a **web of interacting subsystems**
 - cognition
 - group psychology
 - institutions
 - cultural values
 - historical dynamics

Key structure:

- The **field provides a shared state space**
- The **web defines interaction pathways between subsystems**
- Dynamics emerge from feedback across both

A hybrid system: continuous field + discrete interaction web.

Core Summary Statement

Human societies can be modeled as a **dynamic, multi-layer web of interacting systems embedded in a continuous cognitive trait manifold**, where:

- population-level cognitive variation forms a shifting probability landscape,
- social values and group psychology act as feedback mechanisms shaping that landscape,
- and system stability emerges from the relative balance (ratios) of manifold regions rather than fixed categories,
- with instability leading to rigidity, collapse, or phase transitions into new equilibria.

Additions

MASTER LIST 1 — CMTR MANIFOLDS

1. Systematizing Manifold

The tendency and capacity to perceive, decompose, quantify, model, predict, manipulate, optimize, and construct systems.

Micro-Systematizing

Bottom-up analysis.

- decomposition
- measurement
- quantification
- variable isolation
- rule identification
- mechanism analysis
- causal modeling
- prediction through component analysis
- optimization and intervention

Core question:

How do the components produce the system?

Macro-Systematizing

Top-down structural analysis.

- whole-system perception
- emergent pattern recognition
- architecture detection
- recursive and multilevel modeling
- structural analogy
- synthesis
- prediction from system dynamics
- construction of new theoretical architectures

Core question:

What is the structure of the whole, and what must its components be doing for that structure to exist?

Important overlap

Systematizing has a particularly important overlap with **Vigilance & Preparedness** and **Agency & Directing**.

A shared sequence may be:

uncertainty → modeling → prediction → intervention/control

The manifolds diverge according to the purpose of the operation:

- **Systematizing:** understand the system.
- **Vigilance & Preparedness:** reduce uncertainty and threat.
- **Agency & Directing:** direct the system toward a desired outcome.

Micro-systematizing may have especially strong overlap because decomposition, measurement, variable isolation, and prediction can all become mechanisms for controlling outcomes.

2. Vigilance & Preparedness Manifold

The tendency toward threat detection, uncertainty sensitivity, vigilance, anticipatory modeling, checking, avoidance, neutralization, rumination, and attempts to reduce uncertainty.

Core dynamics:

- threat prediction
- uncertainty intolerance
- hypervigilance
- checking
- reassurance seeking
- compulsive neutralization
- avoidance
- anticipatory modeling
- catastrophic simulation
- need for certainty

Core orientation:

I need to know that the system is safe.

The overlap with Agency & Directing emerges when anxiety about uncertain outcomes produces an increasing need to constrain, regulate, or eliminate uncontrolled variables.

3. Reflective Introspection Manifold

The tendency toward negative affect, pessimistic prediction, loss sensitivity, dysphoria, hopelessness, withdrawal, rejection sensitivity, rumination, and negative future modeling.

Core dynamics:

- negative prediction bias
- loss sensitivity
- hopelessness
- dysphoria
- withdrawal
- low reward expectation
- rejection sensitivity
- rumination
- affective threat processing

Core orientation:

The future is likely to contain loss, danger, failure, or pain.

4. State Specialization Manifold

The tendency toward compartmentalization, altered integration between memory, consciousness, identity, perception, emotion, and self-states.

Core dynamics:

- compartmentalization
- dissociative amnesia
- depersonalization
- derealization
- identity-state differentiation
- altered state-dependent access
- discontinuities of self-experience
- protective disconnection

This manifold may permit radically different representations of the same person, event, or reality to operate in different states or contexts without full integration.

5. Agency & Directing Manifold

The tendency to regulate vulnerability, uncertainty, relationships, or desired outcomes through dominance, coercion, possession, hierarchy, manipulation, intimidation, retaliation, exploitation, or direct control.

Core dynamics:

- dominance
- possession
- coercion
- hierarchy
- status protection
- manipulation
- intimidation
- retaliation
- instrumentalization
- exploitation
- elimination of unwanted autonomy
- regulation of vulnerability through control

Core orientation:

I need the system to behave according to my desired outcome.

High-risk overlap

The intersection:

Systematizing × Vigilance & Preparedness × Agency & Directing

may form a particularly important high-risk/high-reward region.

- Systematizing supplies the ability to understand the system.
- Vigilance & Preparedness supplies pressure to reduce uncertainty.
- Agency & Directing supplies motivation to direct outcomes.

At an adaptive extreme:

- scientist
- diagnostician
- investigator
- engineer
- strategist
- systems designer

At a maladaptive extreme:

- compulsive controller
- coercive strategist
- instrumentalizer
- sophisticated antisocial system manipulator

The danger increases when **instrumental/systematizing intelligence becomes decoupled from moral intelligence, autonomy recognition, empathy, and epistemic humility.**

6. Compulsive Engagement Manifold

The tendency toward reinforcement seeking, craving, compulsive repetition, escalation, reward dysregulation, and difficulty disengaging from reinforcing stimuli or behaviors.

Core dynamics:

- reward sensitivity
 - reinforcement seeking
 - craving
 - compulsive repetition
 - escalation
 - impulse/reward dysregulation
 - habit formation
 - difficulty disengaging
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7. Salience Restructuring Manifold

A broad manifold involving altered salience, affective activation and deactivation, unusual cognitive or perceptual processing, psychosis-spectrum phenomena, mood cycling, grandiosity, reality-model disruption, and cognitive disorganization.

Core dynamics may include:

- altered salience
- unusual perceptual processing
- psychosis-spectrum cognition
- mood activation
- mood deactivation
- cycling

- grandiosity
 - cognitive disorganization
 - reality-model disruption
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8. Interpersonal Prediction Manifold

The broad field governing social perception, interpersonal modeling, attachment, emotional perception, emotional regulation, reciprocity, communication, belonging, relational prediction, and social processing.

Core dynamics:

- social modeling
- interpersonal prediction
- attachment
- emotional perception
- emotional regulation
- reciprocity
- communication
- belonging
- relational processing

Important distinction:

Social difficulty does **not** necessarily equal low social intelligence.

A person may accurately perceive a social system while refusing to conform to it.

9. Epistemic Orientation Manifold

A neutral manifold governing the relationship between independently generated judgment and the surrounding social system.

Conformity pole

- consensus sensitivity
- norm adherence
- authority alignment
- affiliation-driven updating
- social-role stability

- discomfort with social disapproval
- tendency to defer to collective reality models

Autonomy pole

- independent judgment
- self-authorship
- norm evaluation
- resistance to coercive consensus
- epistemic independence
- tolerance of social disapproval
- willingness to maintain a model that conflicts with social consensus

Neither pole is inherently good or bad.

High conformity can support:

- coordination
- cooperation
- cultural continuity
- rapid collective action

High autonomy can support:

- innovation
- dissent
- independent error correction
- resistance to harmful systems

Both can become maladaptive in extreme configurations.

MASTER LIST 2 — CMTR INTELLIGENCES

1. Micro-Systematizing Intelligence

The capacity for bottom-up structural reasoning.

Core question:

How do these components produce the whole?

Abilities include:

- decomposition
 - measurement
 - variable analysis
 - mechanism identification
 - quantitative reasoning
 - causal modeling
 - debugging
 - procedural construction
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2. Macro-Systematizing / Creative / Imaginative Intelligence

The capacity for top-down structural reasoning, whole-system modeling, synthesis, analogy, and generation of new architectures.

Core question:

What is the whole structure, and what must its components be doing for it to exist?

Abilities include:

- structural synthesis
 - cross-domain analogy
 - emergent pattern detection
 - recursive modeling
 - theory construction
 - creative recombination
 - architectural reasoning
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3. Intuitive Intelligence

Rapid, largely nonconscious pattern recognition and whole-pattern apprehension preceding conscious sequential reasoning.

Abilities include:

- rapid synthesis
 - implicit pattern detection
 - predictive intuition
 - nonconscious model generation
 - immediate structural recognition
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4. Social Intelligence

The capacity to accurately perceive and model:

- individuals
- motives
- relationships
- groups
- institutions
- hierarchies
- norms
- incentives
- narratives
- collective behavior

High social intelligence does not imply high conformity.

A person may understand a social system accurately and reject it.

5. Emotional Intelligence

Ontological Emotional Intelligence

Understanding how emotional systems work.

Examples:

- recognizing emotional states
- understanding emotional causation
- predicting emotional behavior
- modeling emotional systems

Applied Emotional Intelligence

The ability to deploy emotional understanding in real interaction.

Applied emotional intelligence may be:

- reciprocal/prosocial
- neutral
- instrumental
- manipulative
- Agency & Directing-oriented

Therefore:

High emotional intelligence does not automatically imply high empathy or high morality.

6. Moral Intelligence

The capacity to understand and evaluate:

- harm
- autonomy
- rights
- reciprocity
- justice
- responsibility
- consequences
- obligations
- legitimacy of rules
- independent value of other subjects
- the difference between what can be done and what should be done

A crucial civilizational function of moral intelligence is preventing advanced systematizing capacity from collapsing into pure instrumental optimization.

7. Kinesthetic / Somatic Intelligence

The capacity for embodied cognition and physical information processing.

Abilities include:

- movement

- coordination
 - reflexes
 - bodily perception
 - spatial action
 - sensorimotor prediction
 - physical problem-solving
 - embodied adaptation
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8. Instinctual Intelligence

Rapid nonconscious processing of environmental, bodily, interpersonal, and threat-related information.

Abilities include:

- threat detection
 - environmental reading
 - rapid behavioral prediction
 - nonconscious decision processing
 - instinctive physical response
 - rapid integration of weak signals
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MASTER LIST 3 — MANIFOLDS WITH ALL POSSIBLE DSM DISORDERS BELOW THEM

1. SYSTEMATIZING

Possible membership:

- Autism Spectrum Disorder
- ADHD
- Specific Learning Disorder
- Language Disorder
- Social (Pragmatic) Communication Disorder
- Developmental Coordination Disorder

- OCD
 - OCPD
 - Hoarding Disorder
 - Body Dysmorphic Disorder
 - Schizotypal Personality Disorder
 - Schizophrenia
 - Schizophreniform Disorder
 - Schizoaffective Disorder
 - Brief Psychotic Disorder
 - Delusional Disorder
 - Other/Unspecified Schizophrenia Spectrum and Other Psychotic Disorder
 - Bipolar I Disorder
 - Bipolar II Disorder
 - Cyclothymic Disorder
 - Dissociative Disorders
 - Somatic Symptom Disorder
 - Illness Anxiety Disorder
 - Functional Neurological Symptom Disorder
 - Feeding and Eating Disorders
 - Paraphilic Disorders
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2. VIGILANCE & PREPAREDNESS

Possible membership:

- Generalized Anxiety Disorder
- Panic Disorder
- Agoraphobia
- Social Anxiety Disorder
- Specific Phobia
- Separation Anxiety Disorder
- Selective Mutism
- OCD
- Body Dysmorphic Disorder
- Hoarding Disorder
- Trichotillomania
- Excoriation Disorder
- Illness Anxiety Disorder
- Somatic Symptom Disorder
- PTSD
- Acute Stress Disorder
- Adjustment Disorders

- Prolonged Grief Disorder
 - Avoidant Personality Disorder
 - Dependent Personality Disorder
 - OCPD
 - Paranoid Personality Disorder
 - Schizotypal Personality Disorder
 - Borderline Personality Disorder
 - Delusional Disorder
 - Schizophrenia-spectrum Disorders
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3. REFLECTIVE INTROSPECTION

Possible membership:

- Major Depressive Disorder
 - Persistent Depressive Disorder
 - Other/Unspecified Depressive Disorders
 - Premenstrual Dysphoric Disorder
 - Generalized Anxiety Disorder
 - Panic Disorder
 - Agoraphobia
 - Social Anxiety Disorder
 - Adjustment Disorders
 - Prolonged Grief Disorder
 - PTSD
 - Acute Stress Disorder
 - Avoidant Personality Disorder
 - Dependent Personality Disorder
 - Borderline Personality Disorder
 - Paranoid Personality Disorder
 - Schizotypal Personality Disorder
 - Bipolar I Disorder
 - Bipolar II Disorder
 - Cyclothymic Disorder
 - Schizoaffective Disorder
 - Schizophrenia
 - Delusional Disorder
 - Somatic Symptom Disorder
 - Illness Anxiety Disorder
 - Feeding and Eating Disorders
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4. STATE SPECIALIZATION

Possible membership:

- Dissociative Identity Disorder
 - Dissociative Amnesia
 - Depersonalization/Derealization Disorder
 - Other Specified Dissociative Disorder
 - Unspecified Dissociative Disorder
 - PTSD
 - Acute Stress Disorder
 - Borderline Personality Disorder
 - Schizotypal Personality Disorder
 - Schizophrenia-spectrum Disorders
 - Major Depressive Disorder
 - Bipolar I Disorder
 - Bipolar II Disorder
 - Cyclothymic Disorder
 - Functional Neurological Symptom Disorder
 - Somatic Symptom Disorder
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5. AGENCY & DIRECTING

Possible membership:

- Antisocial Personality Disorder
- Narcissistic Personality Disorder
- Borderline Personality Disorder
- Paranoid Personality Disorder
- Histrionic Personality Disorder
- Conduct Disorder
- Oppositional Defiant Disorder
- Intermittent Explosive Disorder
- Other Disruptive, Impulse-Control, and Conduct Disorders
- Paraphilic Disorders
- Sexual Sadism Disorder
- Other Specified Paraphilic Disorder
- Unspecified Paraphilic Disorder
- Substance Use Disorders
- Gambling Disorder
- Kleptomania
- Pyromania

- Delusional Disorder
 - Bipolar Disorders
 - PTSD
 - Schizophrenia-spectrum Disorders
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6. COMPULSIVE ENGAGEMENT

Possible membership:

- Alcohol Use Disorder
 - Cannabis Use Disorder
 - Opioid Use Disorder
 - Stimulant Use Disorder
 - Sedative/Hypnotic/Anxiolytic Use Disorder
 - Tobacco Use Disorder
 - Phencyclidine Use Disorder
 - Other Hallucinogen Use Disorder
 - Inhalant Use Disorder
 - Other/Unknown Substance Use Disorder
 - Gambling Disorder
 - Binge-Eating Disorder
 - Bulimia Nervosa
 - Trichotillomania
 - Excoriation Disorder
 - Kleptomania
 - Pyromania
 - Paraphilic Disorders
 - Sexual Sadism Disorder
 - ADHD
 - Bipolar Disorders
 - Borderline Personality Disorder
 - Antisocial Personality Disorder
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7. SALIENCE RESTRUCTURING

Possible membership:

- Schizophrenia
- Schizoaffective Disorder
- Schizophreniform Disorder

- Brief Psychotic Disorder
 - Delusional Disorder
 - Other/Unspecified Schizophrenia Spectrum and Other Psychotic Disorder
 - Schizotypal Personality Disorder
 - Bipolar I Disorder
 - Bipolar II Disorder
 - Cyclothymic Disorder
 - Other/Unspecified Bipolar and Related Disorder
 - Substance/Medication-Induced Bipolar and Related Disorder
 - Bipolar/Related Disorder Due to Another Medical Condition
 - Substance/Medication-Induced Psychotic Disorder
 - Psychotic Disorder Due to Another Medical Condition
 - Major Depressive Disorder with Psychotic Features
 - Neurocognitive Disorders with Psychotic Features
 - Catatonia
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8. INTERPERSONAL PREDICTION

Possible membership:

- Autism Spectrum Disorder
- Social (Pragmatic) Communication Disorder
- Language Disorder
- Social Anxiety Disorder
- Selective Mutism
- Separation Anxiety Disorder
- Avoidant Personality Disorder
- Dependent Personality Disorder
- Histrionic Personality Disorder
- Narcissistic Personality Disorder
- Borderline Personality Disorder
- Antisocial Personality Disorder
- Paranoid Personality Disorder
- Schizotypal Personality Disorder
- OCPD
- Conduct Disorder
- Oppositional Defiant Disorder
- Reactive Attachment Disorder
- Disinhibited Social Engagement Disorder
- PTSD
- Acute Stress Disorder
- Adjustment Disorders

- Prolonged Grief Disorder
 - Major Depressive Disorder
 - Persistent Depressive Disorder
 - Bipolar Disorders
 - Schizoaffective Disorder
 - Schizophrenia
 - Delusional Disorder
 - Somatic Symptom Disorder
 - Illness Anxiety Disorder
 - Feeding and Eating Disorders
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9. EPISTEMIC ORIENTATION

Potentially relevant to every DSM disorder.

Especially relevant possible configurations include:

- Autism Spectrum Disorder
 - ADHD
 - OCPD
 - Paranoid Personality Disorder
 - Schizotypal Personality Disorder
 - Antisocial Personality Disorder
 - Narcissistic Personality Disorder
 - Borderline Personality Disorder
 - Avoidant Personality Disorder
 - Dependent Personality Disorder
 - Conduct Disorder
 - Oppositional Defiant Disorder
 - PTSD
 - Dissociative Disorders
 - Bipolar Disorders
 - Schizophrenia-spectrum Disorders
 - OCD
 - Anxiety Disorders
 - Depressive Disorders
 - Substance Use Disorders
 - Feeding and Eating Disorders
 - Paraphilic Disorders
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MASTER LIST 4 — MANIFOLDS WITH DISORDERS DIVIDED INTO DOMINANT / SUB-DOMINANT / POSSIBLE

1. SYSTEMATIZING

Dominant

- Autism Spectrum Disorder
- ADHD
- Specific Learning Disorder
- OCD
- OCPD
- Hoarding Disorder
- Schizotypal Personality Disorder
- Schizophrenia-spectrum Disorders
- Delusional Disorder
- Certain Bipolar configurations

Sub-dominant

- Dissociative Disorders
- PTSD
- Body Dysmorphic Disorder
- Somatic Symptom Disorder
- Illness Anxiety Disorder
- Functional Neurological Symptom Disorder
- Feeding and Eating Disorders
- Paraphilic Disorders
- Borderline Personality Disorder
- Paranoid Personality Disorder
- Bipolar Disorders
- Schizoaffective Disorder

Possible

All other disorders depending on the individual's cognitive architecture.

2. VIGILANCE & PREPAREDNESS

Dominant

- OCD
- Generalized Anxiety Disorder
- Panic Disorder
- Agoraphobia
- Social Anxiety Disorder
- Specific Phobia
- Separation Anxiety Disorder
- Selective Mutism
- Body Dysmorphic Disorder
- Hoarding Disorder
- Trichotillomania
- Excoriation Disorder
- Illness Anxiety Disorder
- Avoidant Personality Disorder
- Dependent Personality Disorder
- OCPD
- Paranoid Personality Disorder
- Schizotypal Personality Disorder

Sub-dominant

- PTSD
- Acute Stress Disorder
- Adjustment Disorders
- Prolonged Grief Disorder
- Borderline Personality Disorder
- Major Depressive Disorder
- Persistent Depressive Disorder
- Bipolar Disorders
- Delusional Disorder
- Schizophrenia-spectrum Disorders
- Functional Neurological Symptom Disorder
- Feeding and Eating Disorders

Possible

All remaining disorders.

3. REFLECTIVE INTROSPECTION

Dominant

- Major Depressive Disorder
- Persistent Depressive Disorder
- Other Depressive Disorders
- Generalized Anxiety Disorder
- Panic Disorder
- Agoraphobia
- Social Anxiety Disorder
- Adjustment Disorders
- Prolonged Grief Disorder
- PTSD
- Acute Stress Disorder
- Avoidant Personality Disorder
- Dependent Personality Disorder
- Borderline Personality Disorder
- Paranoid Personality Disorder
- Schizotypal Personality Disorder
- Anorexia Nervosa
- Bulimia Nervosa
- Binge-Eating Disorder
- ARFID

Sub-dominant

- Bipolar I Disorder
- Bipolar II Disorder
- Cyclothymic Disorder
- Schizoaffective Disorder
- Schizophrenia
- Delusional Disorder
- Somatic Symptom Disorder
- Illness Anxiety Disorder
- OCD
- Substance Use Disorders
- Feeding and Eating Disorders

Possible

All remaining disorders.

4. STATE SPECIALIZATION

Dominant

- Dissociative Identity Disorder
- Dissociative Amnesia
- Depersonalization/Derealization Disorder
- Other Specified Dissociative Disorder
- Unspecified Dissociative Disorder
- PTSD with substantial dissociative organization
- Acute Stress Disorder with substantial dissociative organization
- Schizotypal Personality Disorder

Sub-dominant

- Borderline Personality Disorder
- Schizophrenia-spectrum Disorders
- Schizoaffective Disorder
- Brief Psychotic Disorder
- Delusional Disorder
- Major Depressive Disorder
- Bipolar I Disorder
- Bipolar II Disorder
- Cyclothymic Disorder
- Functional Neurological Symptom Disorder
- Somatic Symptom Disorder
- PTSD without dominant dissociative architecture

Possible

All remaining disorders depending on individual dissociative traits.

5. AGENCY & DIRECTING

Dominant

- Antisocial Personality Disorder
- Narcissistic Personality Disorder
- Conduct Disorder
- Oppositional Defiant Disorder
- Intermittent Explosive Disorder

- Paraphilic Disorders centered on dominance/coercion
- Sexual Sadism Disorder
- Other Specified Paraphilic Disorder involving dominance/control

Sub-dominant

- Borderline Personality Disorder
- Paranoid Personality Disorder
- Histrionic Personality Disorder
- Kleptomania
- Pyromania
- Gambling Disorder
- Substance Use Disorders
- Bipolar Disorders
- Delusional Disorder
- Schizophrenia-spectrum Disorders
- PTSD
- Acute Stress Disorder

Possible

All remaining disorders.

6. COMPULSIVE ENGAGEMENT

Dominant

- Substance Use Disorders
- Gambling Disorder
- Kleptomania
- Pyromania
- Trichotillomania
- Excoriation Disorder
- Binge-Eating Disorder
- Bulimia Nervosa
- Paraphilic Disorders with strong reinforcement/compulsive architecture
- Sexual Sadism Disorder where reinforcement is central

Sub-dominant

- ADHD
- OCD

- Bipolar I Disorder
- Bipolar II Disorder
- Cyclothymic Disorder
- Borderline Personality Disorder
- Antisocial Personality Disorder
- Narcissistic Personality Disorder
- PTSD
- Feeding and Eating Disorders
- Paraphilic Disorders

Possible

All remaining disorders.

7. SALIENCE RESTRUCTURING

Dominant

- Schizophrenia
- Schizoaffective Disorder
- Schizophreniform Disorder
- Brief Psychotic Disorder
- Delusional Disorder
- Other/Unspecified Schizophrenia Spectrum and Other Psychotic Disorder
- Schizotypal Personality Disorder
- Bipolar I Disorder
- Bipolar II Disorder
- Cyclothymic Disorder
- Other Bipolar and Related Disorders
- Substance/Medication-Induced Bipolar and Related Disorder
- Bipolar/Related Disorder Due to Another Medical Condition
- Substance/Medication-Induced Psychotic Disorder
- Psychotic Disorder Due to Another Medical Condition

Sub-dominant

- Major Depressive Disorder with Psychotic Features
- Other depressive disorders with psychotic features
- Borderline Personality Disorder
- Paranoid Personality Disorder
- PTSD with substantial reality/perception disruption
- Dissociative Disorders with substantial reality-integration disturbance

Possible

All remaining disorders.

8. INTERPERSONAL PREDICTION

Dominant

- Social Anxiety Disorder
- Autism Spectrum Disorder in social/emotional-processing dimensions
- Social (Pragmatic) Communication Disorder
- Reactive Attachment Disorder
- Disinhibited Social Engagement Disorder
- Avoidant Personality Disorder
- Dependent Personality Disorder
- Histrionic Personality Disorder
- Narcissistic Personality Disorder
- Borderline Personality Disorder
- Antisocial Personality Disorder
- Paranoid Personality Disorder
- Schizotypal Personality Disorder

Sub-dominant

- PTSD
- Acute Stress Disorder
- Adjustment Disorders
- Prolonged Grief Disorder
- Major Depressive Disorder
- Persistent Depressive Disorder
- Bipolar Disorders
- Schizoaffective Disorder
- Schizophrenia
- Delusional Disorder
- Somatic Symptom Disorder
- Illness Anxiety Disorder
- OCD
- OCPD
- Feeding and Eating Disorders

Possible

All remaining disorders.

9. EPISTEMIC ORIENTATION

Dominant

No DSM diagnosis inherently determines a dominant position on this manifold.

Both high-conformity and high-autonomy configurations may occur within the same diagnosis.

High-conformity configurations may be especially relevant in:

- OCPD
- Dependent Personality Disorder
- Avoidant Personality Disorder
- Social Anxiety Disorder
- Certain PTSD configurations
- Certain Autism Spectrum configurations
- Certain Vigilance & Preparedness configurations
- Certain depressive configurations

High-autonomy configurations may be especially relevant in:

- Oppositional Defiant Disorder
- Conduct Disorder
- Antisocial Personality Disorder
- Certain Autism Spectrum configurations
- Certain ADHD configurations
- Certain Schizotypal configurations
- Certain Paranoid configurations
- Certain Bipolar configurations

Sub-dominant

Potentially relevant to virtually every disorder.

Possible

Every DSM disorder.

Note: Here is the same list without stigmatizing diagnostic language, below:

MASTER LIST 4 — CMTR MANIFOLDS WITH TRAITS DIVIDED INTO BENEFITS, TRADEOFFS, AND ENVIRONMENTAL FIT

The nine CMTR manifolds remain:

1. **Systematizing** — with Micro- and Macro-Systematizing as distinct expressions of the same manifold
2. **Vigilance & Preparedness**
3. **Reflective Introspection**
4. **State Specialization**
5. **Agency & Directing**
6. **Compulsive Engagement**
7. **Salience Restructuring**
8. **Interpersonal Prediction**
9. **Epistemic Orientation**

The categories below describe **trait configurations**, not diagnoses. A trait appearing under "tradeoffs" is not inherently negative; it is a characteristic whose value depends strongly on context, intensity, interaction with other traits, and environmental demands.

1. SYSTEMATIZING MANIFOLD

The Systematizing Manifold contains both **Micro-Systematizing** and **Macro-Systematizing**.

They are **not separate manifolds**.

Many individuals possess substantial traits from **both** expressions, and can move between them depending on the problem, scale, and environment.

1A. MICRO-SYSTEMATIZING

Benefits

- decomposition of complex problems into components
- precise analysis
- measurement
- quantification
- variable isolation
- mechanism identification
- causal analysis
- debugging
- error detection
- procedural reasoning
- sequential reasoning
- precision
- consistency
- detail sensitivity
- rule identification
- technical analysis
- mathematical reasoning
- computational reasoning
- optimization
- controlled experimentation
- systematic comparison
- reproducibility
- quality control
- anomaly detection
- procedural construction
- incremental refinement
- identifying exactly where a system fails
- identifying which variable is responsible for an outcome
- translating complex systems into manageable components
- maintaining attention to fine-grained structural relationships

Tradeoffs

- may prioritize component-level accuracy over emergent behavior
- may become excessively detail-oriented when a problem requires abstraction
- may spend substantial cognitive resources resolving low-level discrepancies
- may prefer known variables over poorly specified variables
- may become frustrated by imprecise instructions
- may overanalyze when rapid approximation would suffice
- may continue debugging after the practically important problem has already been solved
- may have difficulty accepting "good enough" solutions when precision matters less than speed
- may become overly attached to internally coherent procedures
- may underweight variables that are difficult to quantify

- may preferentially trust analyzable information over ambiguous contextual information

Particularly adaptive environments

- engineering
- software development
- mathematics
- laboratory research
- data analysis
- accounting
- auditing
- diagnostics
- quality assurance
- technical troubleshooting
- architecture at the component level
- electronics
- mechanics
- programming
- manufacturing
- experimental science
- forensic analysis
- technical medicine
- any environment requiring precise identification of mechanisms or component failures

Particularly maladaptive environments

- environments requiring constant broad-scale improvisation with little opportunity for component analysis
 - situations in which precision is consistently less valuable than speed
 - environments where variables cannot reasonably be isolated
 - highly ambiguous social environments in which explicit causal structure is unavailable
 - contexts where exhaustive analysis prevents timely action
 - systems whose most important properties are primarily emergent and cannot be understood from component analysis alone
-

1B. MACRO-SYSTEMATIZING

Macro-Systematizing is **whole-system, top-down, emergent, recursive, and dynamically adaptive system perception.**

It is particularly important to distinguish this from the stereotype of "being distracted."

A macro-systematizer may rapidly **filter out information that is not currently relevant to the system-level model**, while simultaneously tracking multiple high-value variables.

What an observer may misclassify as "distractibility" can therefore be **rapid filtering**.

Benefits

- whole-system perception
- emergent pattern recognition
- architectural reasoning
- structural synthesis
- recursive reasoning
- multilevel modeling
- rapid identification of the variables that matter most
- rapid filtering of low-value information
- rapid integration of changing information
- recognition of changing system states
- prediction of system-level consequences
- structural analogy
- cross-domain transfer
- abstraction
- synthesis
- theory construction
- conceptual compression
- recognizing relationships between apparently unrelated phenomena
- generating new architectures
- identifying hidden system constraints
- recognizing feedback loops
- recognizing phase transitions
- recognizing cascading effects
- updating models rapidly when conditions change
- maintaining multiple interacting variables simultaneously
- functioning effectively when information is incomplete
- functioning effectively when the correct procedure is not known in advance
- rapid improvisational problem solving
- remaining cognitively flexible when circumstances change
- maintaining broad situational awareness
- detecting the emergence of new patterns before they become obvious
- rapidly determining which information deserves attention
- **remaining unusually calm and cognitively organized under high adrenaline or acute pressure**
- **rapid response to novel, high-stress, changing conditions**
- **rapid restructuring of the working model when the environment changes**

- **high performance in emergencies where established procedures are insufficient or conditions change faster than procedures can be followed**

This makes Macro-Systematizing particularly relevant to environments involving:

- emergency medicine
- emergency management
- search and rescue
- firefighting
- disaster response
- military command and rapidly changing tactical environments
- emergency aviation
- air-traffic crisis management
- crisis engineering
- incident response
- emergency dispatch
- acute technical failure
- crisis leadership
- rapidly changing scientific environments
- investigative environments where the relevant structure is initially unknown

The key capacity is not simply "thinking quickly."

It is:

rapidly reconstructing the model of the system as the system changes.

Tradeoffs

- may rapidly discard information that appears irrelevant but is later revealed to matter
- may become impatient with environments requiring slow sequential processing
- may jump between levels of abstraction
- may produce conclusions that are difficult for others to reconstruct step-by-step
- may appear inattentive when actually filtering aggressively
- may lose interest once the underlying architecture has become obvious
- may have difficulty maintaining engagement with repetitive low-information tasks
- may prioritize emerging structural information over established procedure
- may challenge procedures that appear mismatched to current conditions
- may underperform when forced to process every available detail equally
- may become frustrated when others require explicit sequential explanation of an already-intuitive structure
- may generate models faster than they can communicate them
- may rapidly revise hypotheses, which can appear inconsistent to observers who do not see the underlying model updates

Particularly adaptive environments

- emergency medicine
- trauma medicine
- emergency nursing
- paramedicine
- firefighting
- search and rescue
- disaster response
- emergency management
- emergency dispatch
- military command
- aviation emergencies
- air-traffic crisis management
- crisis engineering
- cybersecurity incident response
- rapidly evolving technical operations
- investigative work
- intelligence analysis
- complex negotiations
- crisis leadership
- entrepreneurship under uncertainty
- scientific discovery
- theoretical research
- strategic planning
- systems architecture
- complex organizational troubleshooting
- any environment in which **novel information arrives rapidly and the underlying system must be reconstructed in real time**

Particularly maladaptive environments

- highly repetitive work with little structural variation
- environments requiring continuous attention to every low-level detail regardless of relevance
- rigid procedural environments where deviation is unacceptable even when circumstances change
- tasks in which the correct response is always predetermined and no system-level updating is required
- environments that punish rapid hypothesis revision
- contexts where filtering information is mistaken for failure to attend
- situations where broad structural reasoning is consistently subordinated to exhaustive procedural compliance

Micro + Macro Systematizing

Many people possess **both**.

For example, an individual might:

- use Macro-Systematizing to identify the architecture of a failing system
- use Micro-Systematizing to isolate the precise component responsible
- return to Macro-Systematizing to understand the broader consequences
- return to Micro-Systematizing to implement the correction

The two expressions therefore form a **recursive loop**:

Macro → Micro → Macro → Micro

Macro-systematizing identifies the structure.

Micro-systematizing determines the mechanism.

Macro-systematizing determines what that mechanism means for the whole.

Micro-systematizing determines how to alter it precisely.

A person can therefore be high in both, high primarily in one, or display a shifting balance depending on context.

2. VIGILANCE & PREPAREDNESS MANIFOLD

Vigilance & Preparedness — Benefits

- threat detection
- anomaly detection
- vigilance
- anticipation of potential problems
- **preparedness**
- contingency planning
- risk identification
- safety planning

- error detection
- responsibility sensitivity
- prevention-oriented thinking
- thoroughness
- caution
- anticipatory modeling
- identification of vulnerabilities
- preparation for adverse scenarios

Particularly adaptive environments

- safety engineering
- risk assessment
- auditing
- quality control
- cybersecurity
- compliance
- aviation safety
- disaster **planning and preparedness**
- emergency preparedness
- preventive medicine
- investigative work
- environments where anticipating potential failure **before it occurs** is valuable

Particularly maladaptive environments

- acute emergency response requiring rapid action under changing conditions
- environments requiring calm, rapid updating during unpredictable events
- situations where excessive threat appraisal interferes with decision-making
- environments requiring immediate commitment despite incomplete information
- contexts where exhaustive checking or uncertainty reduction delays action

3. REFLECTIVE INTROSPECTION MANIFOLD

Benefits

- sensitivity to loss
- realistic recognition of negative possibilities
- caution

- humility
- awareness of failure modes
- sensitivity to interpersonal harm
- recognition of suffering
- careful consideration of consequences
- reflection
- introspection
- awareness of limitations
- sensitivity to exclusion or rejection
- capacity to detect problems others overlook

Tradeoffs

- pessimistic forecasting
- reduced reward expectation
- withdrawal
- rumination
- self-criticism
- reduced initiative
- heightened loss sensitivity
- rejection sensitivity
- reduced confidence in positive outcomes
- difficulty sustaining motivation
- tendency to overweight negative evidence

Particularly adaptive environments

- risk assessment
- critical review
- ethics
- investigative work
- environments where identifying failure is important
- caregiving contexts requiring sensitivity to suffering
- reflective or analytical environments
- situations requiring careful consideration of adverse consequences

Particularly maladaptive environments

- environments requiring sustained optimism and initiative
 - highly competitive environments demanding repeated rejection tolerance
 - contexts where rapid action is more valuable than prolonged negative forecasting
 - environments in which withdrawal prevents corrective feedback
-

4. STATE SPECIALIZATION MANIFOLD

Benefits

- compartmentalization
- emotional distancing when necessary
- capacity to function during overwhelming circumstances
- state-dependent specialization
- psychological distance from overwhelming information
- ability to maintain functioning under extreme stress
- separation of incompatible information
- absorption
- imaginative immersion
- internal complexity
- capacity to operate through distinct cognitive/emotional states
- temporary reduction of overwhelming affect
- ability to continue performing despite acute emotional overload

Tradeoffs

- discontinuity of memory
- reduced integration between emotional and cognitive information
- difficulty accessing information across states
- depersonalization
- derealization
- emotional disconnection
- compartmentalization becoming rigid
- fragmented autobiographical continuity
- difficulty maintaining a unified self-model
- information remaining isolated in separate processing states

Particularly adaptive environments

- acute crisis environments
- situations requiring temporary emotional compartmentalization
- emergency response
- high-intensity caregiving
- environments requiring functioning during overwhelming events
- creative environments involving deep absorption
- contexts requiring temporary separation of emotionally incompatible tasks

Particularly maladaptive environments

- environments requiring continuous autobiographical integration
 - intimate relationships requiring sustained emotional accessibility
 - situations where isolated information must be integrated across contexts
 - environments where memory continuity is essential
 - contexts in which emotional compartmentalization prevents recognition of important information
-

5. AGENCY & DIRECTING MANIFOLD

Agency & Directing is a **capacity manifold**, not inherently an antisocial one.

Its underlying capacity includes:

- agency
- influence
- outcome direction
- leadership
- strategic control
- boundary enforcement
- resource coordination
- organization
- decisiveness
- hierarchy navigation
- intervention
- protection
- resistance to coercion

The important divergence is how that capacity is organized.

Independent Agency and Moral Recognition

The fundamental distinction between prosocial and antisocial configurations of the Agency & Directing manifold is **recognition of other agents as independent subjects**.

Power, influence, dominance, strategic capacity, assertiveness, and the ability to direct outcomes are not inherently antisocial. A person may possess exceptionally high Agency & Directing capacity while simultaneously recognizing that other people possess:

- independent agency

- independent preferences
- independent interests
- personal boundaries
- rights
- autonomy
- moral standing
- the capacity to determine their own states and actions

In a prosocial configuration, these independent states constitute genuine constraints upon the legitimate use of power.

The central principle is therefore:

Other people are not merely components of the system being controlled; they are autonomous agents within the system whose independent states possess intrinsic moral significance.

Antisocial Configuration

The antisocial configuration becomes more likely when an individual does not adequately recognize other people as independent moral subjects, or does not assign their independent interests sufficient weight to constrain the individual's actions.

Other people may instead be represented primarily as:

- resources
- obstacles
- variables
- instruments
- possessions
- competitors
- threats
- extensions of the self
- objects to be manipulated
- objects whose autonomy may legitimately be overridden

The resulting decision architecture can be represented as:

Other person's independent state → treated as an obstacle or variable to be optimized

rather than:

Other person's independent state → incorporated as an independently valuable constraint upon action.

This distinction is particularly important because **antisocial Agency & Directing does not require low social or emotional intelligence.**

An individual may possess extremely sophisticated abilities to:

- perceive social dynamics
- infer motives
- predict emotional responses
- identify vulnerabilities
- model relationships
- anticipate behavior
- manipulate incentives
- understand institutional structures

while simultaneously failing to treat the people being modeled as autonomous moral subjects.

Consequently:

Understanding another person and recognizing another person's autonomy are distinct capacities.

High social or emotional intelligence can therefore be incorporated into either a prosocial or antisocial Agency & Directing configuration.

Prosocial Configuration

In a prosocial configuration:

Power + social/emotional intelligence + moral intelligence + autonomy recognition

can produce:

- protection
- leadership
- coordination
- advocacy
- institutional reform
- crisis management
- boundary enforcement
- prevention of exploitation
- defense of vulnerable agents
- expansion of other people's capacity for self-determination

The underlying principle is:

“I have the capacity to influence the system, but the other agents within it remain autonomous.”

Power is therefore used to **expand, protect, coordinate, or restore the possibility space available to other agents.**

Antisocial Configuration

In an antisocial configuration:

Power + social/emotional intelligence + systematizing intelligence – sufficient moral recognition of independent agency

can produce sophisticated forms of:

- manipulation
- coercion
- exploitation
- domination
- instrumentalization
- possession
- intimidation
- retaliation
- autonomy suppression

The underlying principle becomes:

“I understand the other agents sufficiently well to make them conform to the state I want.”

The problem is therefore not the ability to model other people accurately. It is the failure to incorporate their **independent existence as morally constraining.**

This produces a critical CMTR distinction:

Social intelligence answers: “What is this person thinking, feeling, wanting, and likely to do?”

Moral intelligence answers, in part: “What obligations arise from the fact that this person is an independent person whose interests and autonomy matter independently of my objectives?”

These capacities can vary independently.

Accordingly, a person may be:

high social intelligence + high emotional intelligence + high systematizing intelligence + low moral intelligence/autonomy recognition

and consequently possess an exceptionally sophisticated capacity to understand and manipulate social systems while treating the people within those systems instrumentally.

Conversely:

high social intelligence + high emotional intelligence + high systematizing intelligence + high moral intelligence/autonomy recognition

can produce exceptionally sophisticated prosocial Agency & Directing configurations.

Core Distinction

The distinction can therefore be summarized as:

Prosocial Agency & Directing:

“I can direct outcomes without denying the independent agency of the agents affected by those outcomes.”

Antisocial Agency & Directing:

“The independent agency of other agents is subordinate to my desired outcome.”

Thus, within CMTR:

Power is not the pathology. Control is not inherently pathological. Strategic social understanding is not inherently pathological. Even dominance and high agency are not inherently antisocial.

The critical variable is **whether other agents remain morally real as independent subjects within the individual's model of the system.**

Prosocial Agency & Directing

Benefits

- leadership
- protection
- coordination
- advocacy
- decisive action

- boundary enforcement
- crisis management
- institutional reform
- resource mobilization
- defense of vulnerable people
- autonomy protection
- strategic organization
- enforcement of legitimate rules
- prevention of exploitation
- correction of dysfunctional systems
- willingness to confront harmful actors
- high agency
- high responsibility
- strong outcome orientation
- willingness to assume difficult decisions
- ability to act when others hesitate

Especially when combined with **high Moral Intelligence**, this can produce:

power directed toward expanding or protecting the possibility space of other agents.

Tradeoffs

- may become overly interventionist
- may assume responsibility for systems that others could manage themselves
- may have difficulty tolerating preventable disorder
- may become frustrated by passivity
- may become excessively directive
- may overestimate the value of intervention
- may prioritize outcomes over process
- may become controlling when under severe stress
- may have difficulty relinquishing responsibility
- may experience strong discomfort with helplessness

Particularly adaptive environments

- leadership
- emergency management
- crisis response
- advocacy
- institutional reform
- organizational leadership
- medicine
- law

- public administration
- human-rights work
- protective professions
- entrepreneurship
- systems correction
- disaster response
- environments requiring decisive coordination under uncertainty

Particularly maladaptive environments

- environments where unilateral intervention consistently undermines autonomy
 - contexts requiring prolonged deference rather than leadership
 - situations where control over outcomes is structurally impossible
 - environments where excessive intervention creates more instability than it prevents
-

Antisocial Agency & Directing

The **capacity** for Agency & Directing can coexist with high intelligence.

An antisocial configuration is therefore **not equivalent to low social intelligence or low emotional intelligence**.

Indeed, an individual in this configuration may possess:

- high Social Intelligence
- high Ontological Emotional Intelligence
- high strategic intelligence
- high Micro- and/or Macro-Systematizing Intelligence
- high predictive ability
- high interpersonal modeling ability
- high ability to read motives and vulnerabilities
- high ability to anticipate reactions

The critical divergence is **Moral Intelligence and the incorporation of other agents' autonomy into the decision function**.

Benefits / capacities

Even where the overall configuration is antisocial, the underlying capacities can include:

- strategic intelligence
- social perception
- interpersonal prediction

- emotional-state recognition
- persuasive ability
- negotiation ability
- rapid reading of social hierarchies
- strategic planning
- system manipulation
- high decisiveness
- high risk tolerance
- strong agency
- resistance to social pressure
- ability to function under pressure
- high instrumental competence
- sophisticated understanding of incentives
- ability to identify vulnerabilities
- ability to anticipate consequences
- ability to coordinate complex actions

This is precisely why the configuration can become highly consequential when combined with high systematizing intelligence.

Tradeoffs

The characteristic problem is not lack of capability but **how capability is directed**.

Potential tradeoffs include:

- instrumentalization of other people
- diminished recognition of others as independent ends
- entitlement
- coercion
- domination
- exploitation
- possession
- manipulation
- retaliation
- hierarchy preservation
- autonomy suppression
- moral disengagement
- externalization of responsibility
- strategic deception
- vulnerability avoidance
- control fixation
- treating relationships as transactions
- treating other agents primarily as variables in an optimization problem

Particularly adaptive environments

For the **underlying capacities**, environments can reward:

- strategic competition
- negotiation
- high-stakes leadership
- crisis management
- competitive strategy
- intelligence analysis
- complex organizational politics
- entrepreneurship
- adversarial analysis
- high-pressure decision-making
- environments requiring strong resistance to social pressure

The configuration becomes substantially more adaptive when its social and strategic capacities are constrained by **Moral Intelligence, reciprocity, autonomy recognition, and legitimate institutional limits**.

Particularly maladaptive environments

- environments with unrestricted power
 - systems lacking accountability
 - hierarchical environments with weak oversight
 - competitive systems that reward exploitation
 - environments where coercion produces immediate rewards
 - systems that strongly reward domination without incorporating externalities
 - contexts in which other people's autonomy is treated as an obstacle rather than a system variable with intrinsic value
-

6. COMPULSIVE ENGAGEMENT MANIFOLD

Benefits

- strong reward sensitivity
- enthusiasm
- persistence toward rewarding goals
- intense engagement

- motivation
- novelty seeking
- sensation seeking
- rapid reinforcement learning
- strong habit formation
- capacity for sustained pursuit when reinforcement is available
- responsiveness to immediate feedback
- willingness to experiment
- high engagement with intrinsically rewarding activities

Tradeoffs

- compulsive repetition
- difficulty disengaging
- impulsive reward selection
- escalation
- cue sensitivity
- boredom sensitivity
- immediate-reward bias
- reinforcement dependence
- difficulty delaying gratification
- perseveration
- difficulty abandoning rewarding but ineffective strategies

Particularly adaptive environments

- entrepreneurship
- sales
- creative production
- competitive environments
- exploration
- performance environments
- innovation
- rapidly changing environments
- occupations with immediate feedback and variable reinforcement
- activities requiring intense bursts of engagement

Particularly maladaptive environments

- environments dominated by delayed rewards
- highly repetitive low-reward work
- contexts with abundant highly reinforcing but harmful stimuli
- environments where impulse control is continuously required without meaningful feedback

7. SALIENCE RESTRUCTURING MANIFOLD

This manifold concerns atypical salience, association, activation, perception, and reality-model organization—not a simple deficit category.

Benefits

- divergent association
- unusual pattern recognition
- novel conceptual connections
- perceptual sensitivity
- imaginative richness
- unconventional hypotheses
- associative flexibility
- intense ideation
- creative recombination
- unusual perspectives
- high cognitive novelty
- expansive conceptual generation
- sensitivity to weak or unusual signals
- ability to question established models
- unusual approaches to problems

Tradeoffs

- excessive salience attribution
- unstable prioritization
- cognitive disorganization
- difficulty distinguishing signal from noise
- excessive associative connections
- perceptual instability
- extreme activation
- reduced inhibitory control
- difficulty maintaining a stable model under certain conditions
- social or cognitive withdrawal
- overconfidence in internally generated models
- instability across activation states

Particularly adaptive environments

- creative work
- artistic production
- theoretical exploration
- innovation
- speculative research
- environments rewarding unconventional hypotheses
- conceptual design
- fields where novel associations are valuable

Particularly maladaptive environments

- environments requiring extremely stable interpretation of ambiguous information
 - tasks requiring sustained consistency across fluctuating cognitive states
 - situations where inaccurate salience assignment has immediate consequences
 - contexts demanding strict reality-model stability under extreme activation
-

8. INTERPERSONAL PREDICTION MANIFOLD

Benefits

- social perception
- interpersonal modeling
- emotional perception
- emotional attunement
- perspective taking
- attachment
- communication
- reciprocity
- relational prediction
- social learning
- belonging
- conflict recognition
- interpersonal adaptation
- emotional regulation
- empathy
- social pattern recognition
- understanding group dynamics

- understanding motives
- detecting subtle interpersonal changes
- predicting relational consequences

Tradeoffs

- rejection sensitivity
- social approval sensitivity
- interpersonal vigilance
- emotional overload
- difficulty separating one's own emotions from others' emotions
- excessive affiliation needs
- social inhibition
- emotional overinvestment
- conflict avoidance
- interpersonal rumination
- emotional contagion
- excessive concern with relational consequences

Particularly adaptive environments

- teaching
- counseling
- leadership
- negotiation
- diplomacy
- medicine
- management
- caregiving
- collaboration
- mediation
- communication-intensive work
- community organization
- environments requiring nuanced interpersonal prediction

Particularly maladaptive environments

- prolonged social isolation
 - highly adversarial environments
 - contexts with little interpersonal feedback
 - environments where emotional responsiveness is consistently exploited
 - situations requiring extreme emotional detachment without opportunities for regulation
-

9. EPISTEMIC ORIENTATION MANIFOLD

This manifold is explicitly **morally neutral**.

Neither conformity nor autonomy is intrinsically adaptive.

High-Conformity Configuration

Benefits

- coordination
- cooperation
- norm adherence
- institutional stability
- cultural continuity
- rapid collective action
- group cohesion
- predictable social behavior
- role reliability
- trust in established procedures
- willingness to defer when expertise or legitimate authority warrants it
- preservation of institutional knowledge
- reduced coordination costs

Tradeoffs

- susceptibility to social pressure
- reduced independent evaluation
- authority dependence
- consensus overvaluation
- difficulty dissenting
- institutional inertia
- suppression of unconventional ideas
- excessive conflict avoidance
- vulnerability to harmful group norms
- reduced willingness to challenge established assumptions

Particularly adaptive environments

- coordinated emergency response
- aviation

- military operations
- surgical teams
- highly regulated safety environments
- organizations requiring standardized procedures
- contexts where rapid collective coordination is essential

Particularly maladaptive environments

- corrupt institutions
 - rapidly changing environments
 - systems with entrenched errors
 - authoritarian environments
 - scientific environments where dissent is necessary
 - contexts in which existing norms are actively harmful
-

High-Autonomy Configuration

Benefits

- independent judgment
- epistemic independence
- innovation
- dissent
- self-direction
- resistance to coercion
- willingness to challenge authority
- independent problem solving
- unconventionality
- experimentation
- intellectual independence
- error correction
- resistance to groupthink
- autonomy protection
- institutional criticism
- creation of new models
- tolerance of social disapproval
- willingness to maintain an independently derived conclusion

Tradeoffs

- resistance to useful coordination
- unnecessary opposition to authority

- difficulty accepting externally imposed procedures
- reduced group cohesion
- excessive skepticism
- difficulty compromising
- rejection of consensus even when consensus is accurate
- underweighting accumulated institutional knowledge
- increased social friction
- tendency toward unnecessary independence

Particularly adaptive environments

- scientific research
- entrepreneurship
- innovation
- investigative journalism
- institutional reform
- artistic creation
- theoretical work
- whistleblowing
- adversarial analysis
- independent scholarship
- environments requiring error correction
- environments where established consensus may be wrong

Particularly maladaptive environments

- highly coordinated emergency operations
- situations requiring immediate procedural compliance
- environments where individual deviation creates significant collective risk
- tasks in which standardized procedures have been carefully optimized for safety
- contexts where coordination is more important than individual optimization

THE CENTRAL CMTR DISTINCTION

The purpose of this structure is to prevent the manifold itself from being confused with either **pathology or virtue**.

A trait can be:

- beneficial in one environment
- costly in another

- beneficial at one intensity
- costly at another intensity
- beneficial in combination with one manifold
- costly in combination with another

And a characteristic that looks maladaptive from the outside may sometimes actually be an **adaptive capability being misclassified because the observer does not understand what the person is doing.**

The clearest example is Macro-Systematizing:

Rapid filtering is not inherently distractibility.

If a person is rapidly filtering low-value information while continuously updating a high-dimensional model of the environment, the observable behavior may resemble inattentiveness to someone operating with a sequential attentional model.

Likewise:

Rapid model-switching is not inherently inconsistency.

It can represent continuous updating in response to changing system conditions.

And:

Calmness under extreme pressure is not inherently emotional detachment.

For a strong Macro-Systematizer, high arousal can sometimes increase the salience of relevant information while suppressing irrelevant information, producing unusually effective performance in rapidly changing emergencies.

THE NINE-MANIFOLD STRUCTURE

The resulting CMTR structure is therefore:

1. Systematizing

- Micro-Systematizing
- Macro-Systematizing

2. Vigilance & Preparedness

3. Reflective Introspection

4. State Specialization

5. Agency & Directing

- Prosocial configuration
- Antisocial configuration

6. Compulsive Engagement

7. Salience Restructuring

8. Interpersonal Prediction

9. Epistemic Orientation

The important architectural point is that **Micro/Macro** is a subdivision within **Systematizing**, while **prosocial/antisocial** is a configuration distinction within **Agency & Directing**. Neither subdivision creates an additional manifold.

The model can consequently represent a person as a **weighted multidimensional configuration** rather than assigning them a categorical identity:

Systematizing: high

→ Micro: high

→ Macro: very high

State Specialization: moderate-high

Vigilance & Preparedness: moderate

Interpersonal Prediction: high

Agency & Directing: high

→ prosocial configuration: high

Epistemic Orientation: high autonomy

The nine manifolds therefore describe **different dimensions of cognitive/behavioral organization**, not nine classes of people.

1. **Systematizing** — how strongly and in what manner a person models systems
 - Micro: components → mechanisms → whole
 - Macro: whole → architecture → components

2. **Vigilance & Preparedness** — how strongly the system detects and attempts to manage uncertainty and threat
3. **Reflective Introspection** — how strongly the system weights loss, negative prediction, withdrawal, and reduced reward expectation
4. **State Specialization** — how strongly the system compartmentalizes information, affect, memory, perception, or self-state
5. **Agency & Directing** — how strongly the system seeks to direct outcomes, exercise agency, organize systems, or control environments
6. **Compulsive Engagement** — how strongly behavior is organized around reward, reinforcement, repetition, and immediate feedback
7. **Salience Restructuring-related** — how strongly cognition involves altered salience, atypical association, unusual perception, activation, or reality-model variation
8. **Interpersonal Prediction** — how strongly and accurately the system models people, relationships, emotions, and groups
9. **Epistemic Orientation** — how independently generated judgment is positioned relative to collective models

And importantly, **benefits and tradeoffs are not opposites in the sense of one being the “real” trait and the other being pathology**. They are often two consequences of the same underlying architecture under different environmental conditions.

That is particularly important for CMTR:

A trait can be highly adaptive in one environment, costly in another, and potentially catastrophic in a third without the underlying trait itself changing categories.

And the **Agency & Directing manifold makes this especially clear**: high agency, strategic capacity, social intelligence, emotional intelligence, and systematizing intelligence can produce either extraordinarily protective/reformative behavior or extraordinarily sophisticated exploitation depending heavily on the development and strength of moral intelligence, autonomy recognition, reciprocity, and the surrounding reinforcement structure.

FINAL CMTR ENVIRONMENTAL PRINCIPLE

This gives the manifold model a second axis beyond **trait configuration**:

Trait architecture × environmental fit

The same configuration can therefore produce radically different outcomes in different environments.

For example:

High Micro-Systematizing + high precision

can be extraordinarily adaptive in engineering and deeply frustrating in a workplace that requires continuous improvisation.

High Macro-Systematizing + high autonomy

can generate paradigm-changing science in one environment and chronic institutional conflict in another.

High Interpersonal Prediction + high sensitivity

can produce exceptional counseling, teaching, diplomacy, or leadership—or become debilitating in a chronically hostile social environment.

High Agency & Directing + high moral intelligence

can produce extraordinary protective leadership under legitimate accountability—or paternalistic overreach when the person becomes convinced that their own judgment justifies overriding everyone else's.

High Agency & Directing + high social/emotional/systematizing intelligence + low moral/autonomy constraint

can become extremely adaptive **to a predatory environment** while being profoundly maladaptive to a cooperative one.

And:

High Autonomy

can be the mechanism by which a system detects and corrects its own errors—or the mechanism by which an individual refuses necessary coordination.

So the CMTR model does not merely ask:

“What traits does this person have?”

It asks:

“What configuration of traits does this person possess, what capacities and tradeoffs does that configuration generate, and how does the surrounding environment amplify, suppress, reward, or punish those characteristics?”

CORE CMTR MODEL

CMTR operates at four distinct levels:

Intelligence

What kinds of information can the person perceive, model, synthesize, predict, and understand?

Manifold

What recurring configurations organize the person's cognition, affect, behavior, adaptation, and relationship to systems?

Epistemic Orientation

How does the person position independently generated judgment relative to social consensus and surrounding systems?

DSM Diagnosis

What clinically recognizable pattern of symptoms and impairment is present?

The central principle is:

Individuals are not categories. They are locations and trajectories within multiple overlapping manifolds.

A single person may therefore have:

Dominant manifolds:

Systematizing + State Specialization + Autonomy

Sub-dominant manifolds:

Vigilance & Preparedness + Interpersonal Prediction

Secondary or situational manifolds:

Reflective Introspection + Compulsive Engagement

And a different person with the **same DSM diagnosis** may occupy a completely different manifold configuration.

The model is therefore **continuous, overlapping, multidimensional, and dynamic rather than categorical**.

ADDENDUM — PROSOCIAL VS. ANTISOCIAL AGENCY & DIRECTING

The **Agency & Directing manifold** is morally neutral at the level of underlying capacity. The relevant distinction is between **prosocial and antisocial configurations of the manifold**, rather than treating power, influence, agency, or control as intrinsically pathological.

Prosocial Agency & Directing

A prosocial Agency & Directing configuration uses power, agency, influence, and control **in ways that preserve or expand the autonomy and functioning of other systems**.

Examples include:

- protecting vulnerable people
- establishing boundaries
- organizing groups
- coordinating collective action
- leadership
- crisis management
- enforcing legitimate rules
- preventing exploitation
- defending oneself or others
- changing harmful systems
- directing resources toward beneficial outcomes
- creating conditions under which others can exercise greater autonomy

The central distinction is:

Prosocial power increases the capacity of the system and/or protects the autonomy of its constituent agents.

Thus:

Influence can expand possibility space; coercive control contracts it.

A person can therefore be extremely high in Agency & Directing capacity without being antisocial.

Antisocial Agency & Directing

The antisocial configuration emerges when power and control become organized around **instrumentalization, domination, possession, coercion, exploitation, hierarchy maintenance, retaliation, or the suppression of other agents' independent states.**

Its characteristic transformation is:

“I can influence the system” → “The system must conform to my desired state.”

Other people or living systems increasingly become **objects, variables, resources, obstacles, or extensions of the self**, rather than autonomous entities whose independent states have intrinsic importance.

This configuration can become particularly dangerous when combined with:

- high systematizing intelligence
- high predictive ability
- low applied moral intelligence
- low recognition of autonomy
- low empathy or affective empathy
- high reinforcement sensitivity
- strong entitlement
- dissociation/compartmentalization
- low epistemic humility

The resulting system may become extremely capable of **understanding and manipulating other systems without adequately incorporating their independent interests into its decision function.**

DEVELOPMENTAL ORIGIN OF ANTISOCIAL AGENCY & DIRECTING

The antisocial Agency & Directing manifold should generally **not be conceptualized as something a person is simply born with in fully developed form.**

A person may begin with:

- genetic predispositions
- temperament
- unusually high sensitivity
- neurodevelopmental differences
- high threat sensitivity
- high reward sensitivity
- strong systematizing capacity
- unusual emotional intensity
- particular vulnerabilities to environmental conditions

But these predispositions interact recursively with development.

A possible developmental pathway is:

temperament/genetic predisposition

- **developmental environment**
 - repeated emotional states
 - repeated behavioral adaptations
 - reinforcement of those adaptations
 - reorganization of neural circuitry
 - increasingly stable trait architecture
 - Agency & Directing configuration becomes instantiated

In this sense, the antisocial Agency & Directing manifold can be **made through development rather than simply born.**

This does not mean that every person exposed to trauma develops antisociality, nor that developmental trauma is necessary for every antisocial configuration. Rather, **developmental history interacts with predisposition and environmental reinforcement to produce the eventual manifold topology.**

TRAUMA, ENTITLEMENT, AND THE DEVELOPMENT OF AGENCY & DIRECTING

One important developmental distinction is between different forms of early environmental organization.

Severe early threat without sufficient entitlement

Repeated:

- scapegoating
- humiliation
- violence
- unpredictability
- abandonment
- coercion
- helplessness
- chronic threat

may preferentially organize the developing system toward:

State Specialization

and/or

Vigilance & Preparedness

and/or

Reflective Introspection

configurations.

The developing organism learns:

“The world is dangerous; I must monitor, avoid, compartmentalize, predict, or endure.”

Severe early engulfment, overvaluation, and entitlement formation

A different pathway can occur when the child is:

- excessively idealized
- treated as an extension of the parent
- engulfed
- assigned an inappropriate emotional role
- praised for embodying the parent's desired identity
- taught that their desires should govern others
- shielded from normal limits and consequences

- given strong entitlement scripts

This can create:

“My desires should determine the environment.”

When combined with developmental vulnerability, high reward sensitivity, strong dominance tendencies, or later rejection/trauma, this may provide particularly fertile conditions for an **antisocial Agency & Directing configuration**.

Trauma + entitlement

Severe trauma and strong entitlement scripts can also interact.

The resulting developmental equation may become:

“I am entitled to control others because vulnerability means being controlled by them.”

Control then becomes simultaneously:

- defensive
- compensatory
- identity-preserving
- rewarding
- anxiety-reducing
- reinforced through repeated success

Eventually the behavior can cease to feel like a strategy and become part of the person's stable architecture.

EMOTIONAL STATES AS DEVELOPERS OF NEURAL ARCHITECTURE

A central CMTR proposition is therefore:

Emotional states are not merely outputs of the brain; repeated emotional states participate in developing the brain that subsequently produces those states.

A child's repeated experience of:

fear → hypervigilance → prediction → control

can strengthen the neural architecture supporting those processes.

Likewise:

helplessness → dissociation → compartmentalization

can reinforce increasingly efficient dissociative organization.

And:

entitlement → dominance → successful control → reinforcement

can progressively stabilize a Agency & Directing architecture.

The relationship is therefore recursive:

**state → behavior → environment → reinforcement → neural development →
future state**

This creates a developmental feedback loop.

The eventual adult phenotype may therefore look like an intrinsic personality characteristic even though **the architecture was substantially constructed through repeated interaction between predisposition and environment.**

This is one reason CMTR should conceptualize traits as **developmental trajectories through a manifold rather than fixed categories.**

IMPORTANT DISTINCTION: EXPLANATION ≠ EXCULPATION

A developmental account of antisocial Agency & Directing does **not** eliminate responsibility.

The distinction is:

“This is how the configuration developed”

is an explanation of causation.

It is not:

“Therefore the person was not responsible for what they subsequently chose to do.”

Understanding developmental construction is valuable precisely because it identifies **where a maladaptive configuration can potentially be interrupted and reorganized.**

The ultimate objective is not to classify someone as inherently evil or inherently good.

It is to understand:

predisposition → developmental environment → emotional state → adaptation → reinforcement → neural development → trait architecture → behavior

and thereby identify the points at which the trajectory can be altered.

CORE PRINCIPLE

The most important distinction can therefore be stated simply:

Power is not the pathology.

The pathology is the transformation of power into **domination**, of influence into **coercion**, of system understanding into **instrumentalization**, and of agency into **suppression of other agents' autonomy**.

A highly developed Agency & Directing capacity can instead become profoundly prosocial when constrained by:

- moral intelligence
- empathy
- autonomy recognition
- reciprocity
- epistemic humility
- systems awareness
- willingness to accept limits

Thus the same basic capacity for **directing outcomes** can produce radically different trajectories depending on what constrains and organizes it.

Prosocial configuration:

Power → protection → coordination → expanded possibility

Antisocial configuration:

Power → possession → coercion → contracted possibility

And that distinction fits the broader CMTR principle you've established:

Adaptive systems influence their environments while remaining responsive to the independent states of the systems they affect. Maladaptive control attempts to eliminate those independent states.

ADDENDUM — PTSD, C-PTSD, DEVELOPMENTAL TRAUMA, AND MANIFOLD TRAJECTORIES

1. PTSD and C-PTSD Are Not Independent Manifolds

PTSD and C-PTSD should not be represented as independent cognitive or personality manifolds within CMTR.

They are better understood as trauma-response architectures that can develop across any underlying manifold.

Extreme trauma, repetitive trauma, prolonged trauma, or trauma from which escape is impossible can reorganize cognition, affect, perception, memory, behavior, attachment, threat processing, and physiological regulation.

Accordingly:

PTSD/C-PTSD describes what trauma has done to an individual's existing or developing architecture, rather than defining the underlying architecture itself.

A person with a strongly systematizing architecture, social/emotional architecture, dissociative architecture, OCD/anxiety architecture, depressive/anxiety architecture, or Agency & Directing architecture can all develop PTSD.

Trauma therefore acts as a cross-manifold perturbation.

The resulting PTSD/C-PTSD architecture can subsequently feed back into the underlying manifold, sometimes substantially changing its developmental trajectory.

2. Trauma Is Capable of Affecting Any Manifold

The existence of a particular manifold does not determine whether a person can develop PTSD.

Any person can experience trauma.

However, different underlying architectures may produce different responses to the same traumatic environment.

For example:

- A highly threat-sensitive person may develop hypervigilance and compulsive uncertainty reduction.
- A highly dissociative or highly imaginative person may develop extensive compartmentalization.
- A highly systematizing person may respond through increasingly elaborate threat modeling and prediction.
- A socially oriented person may reorganize around attachment, rejection, or interpersonal threat.
- A person with strong Agency & Directing tendencies may respond by attempting to prevent future vulnerability through dominance and control.

Thus:

Trauma is the common developmental pressure; the resulting architecture depends partly upon the structure through which that pressure is processed.

Sensitive neurotypes or particularly vulnerable developmental configurations may therefore be more likely to undergo substantial reorganization under trauma, but trauma is not restricted to any particular neurotype.

3. Childhood Trauma Has a Particularly Strong Relationship With State Specialization Organization

Developmental trauma is especially important because it occurs while the nervous system, attachment system, self-model, emotional regulation system, and social model are still developing.

When trauma is:

- severe
- repetitive
- interpersonal
- unpredictable

- inescapable
- perpetrated by caregivers
- or fundamentally incompatible with the child's dependence upon those caregivers

the developing system may be unable to integrate the experience into a single coherent model of self and reality.

Dissociation can therefore become an adaptive developmental solution.

The developing organism may effectively learn:

"Some experiences cannot be integrated into ordinary conscious functioning while maintaining attachment, safety, or psychological continuity."

This can produce:

- compartmentalization
- state-dependent processing
- dissociative amnesia
- depersonalization
- derealization
- identity-state differentiation
- emotional disconnection
- altered access to memory
- separation of incompatible representations

Consequently:

Childhood developmental trauma strongly predicts a dominant or co-dominant State Specialization organization, although the exact resulting architecture varies among individuals.

The State Specialization manifold can therefore coexist with radically different secondary manifolds.

4. Two Major Developmental Trajectories

A particularly important CMTR distinction is between two broad trajectories that can emerge from developmental trauma.

Trajectory A — Trauma + Entitlement → Agency & Directing

A developing person exposed to severe trauma may nevertheless receive sufficiently powerful entitlement scripts to prevent the trauma from being internalized primarily as:

"I am powerless, defective, or responsible for what is happening to me."

Instead, the developing model may become:

"Other people are supposed to accommodate me, and I must acquire enough power that nobody can ever dominate me again."

The result can be development toward a Agency & Directing-dominant configuration, frequently accompanied by a State Specialization organization.

Possible components include:

- dominance
- entitlement
- externalization of blame
- instrumentalization
- retaliation
- possession
- coercive control
- suppression of vulnerability
- identification with the aggressor
- compartmentalization
- reduced access to empathy under conditions of threat or desire

The trauma remains developmentally important, but the adaptation is organized around power rather than helplessness.

5. Trauma Without Sufficient Entitlement → Internalizing Trajectories

When a traumatized child does not receive sufficiently powerful entitlement scripts, the same developmental pressure may instead be organized around:

"Something is wrong with me."

or:

"I must become better, safer, more compliant, more predictable, or less threatening in order to prevent this from happening again."

This can preferentially produce configurations involving:

- State Specialization organization
- Vigilance & Preparedness
- Reflective Introspection

- avoidance
- social anxiety
- dependency
- hypervigilance
- compulsive self-monitoring
- perfectionism
- self-blame
- withdrawal

The developing person attempts to obtain safety through:

self-modification rather than domination of others.

Thus the two broad adaptations can be represented as:

Trauma + insufficient entitlement

→ "I must change myself to survive."

versus

Trauma + strong entitlement

→ "I must control others/the environment to survive."

Both can coexist with substantial State Specialization organization.

6. Entitlement Scripts Are Not Necessarily Created Inside the Family

Entitlement does not have to originate exclusively in parenting.

A child receives developmental scripts from multiple nested systems:

- family
- peers
- religion
- socioeconomic environment
- institutions
- media
- cultural narratives
- gender roles
- racial or class hierarchies
- national narratives
- historical traditions

Therefore a child can experience severe abuse without being explicitly spoiled or hyperpraised and nevertheless acquire strong cultural scripts telling them that their autonomy, desires, dominance, or authority should take precedence over others'.

Such scripts can change how the child interprets abuse.

Instead of:

"I am bad and deserve this."

the developing model may become:

"The world is supposed to recognize my authority, and the people hurting me are violating the position I am supposed to occupy."

This can increase the probability of externalization, identification with the aggressor, and later Agency & Directing organization.

7. Gendered Entitlement Scripts

Within the CMTR model, culturally transmitted gender-role scripts are therefore potentially important developmental variables.

Historically persistent patriarchal cultures frequently provide boys with stronger explicit or implicit entitlement scripts concerning:

- authority
- dominance
- sexual access
- emotional centrality
- physical aggression
- leadership
- obedience from others
- control of partners and family members

Girls generally receive fewer equivalent culturally sanctioned dominance scripts.

This creates a potential developmental asymmetry:

The same developmental trauma can therefore interact with different culturally available entitlement architectures.

This provides one possible developmental contribution to the observed greater prevalence of antisocial and violent behavior among males.

It does not imply that biological sex mechanically determines a Agency & Directing trajectory.

Rather:

Sex-linked developmental environments alter the probability distribution of available behavioral and identity scripts.

A boy experiencing developmental trauma may therefore have greater access to culturally sanctioned explanations such as:

"I must become stronger than everyone else."

or:

"I must never allow anyone to control me."

A girl experiencing the same trauma may more frequently encounter cultural scripts directing her toward:

"I must become better, more compliant, more attractive, more useful, or less difficult."

These are not universal outcomes. They are culturally mediated probabilities.

8. Hyperpraise, Spoiling, Engulfment, and Enmeshment

Entitlement can also develop through family dynamics that appear superficially positive but interfere with autonomous development.

Excessive:

- hyperpraise
- idealization
- projection
- spoiling
- engulfment
- enmeshment
- emotional parentification
- assigning the child the role of emotional partner
- treating the child as an extension of the parent's identity

can communicate:

"You are extraordinary because you embody what I need you to be."

This is fundamentally different from healthy affirmation.

Healthy affirmation communicates:

"You are valuable as an autonomous person."

Engulfing idealization communicates:

"You are valuable insofar as you fulfill my projected identity for you."

The latter can simultaneously produce:

- grandiosity
- entitlement
- fear of abandonment
- fear of engulfment
- fragile self-esteem
- difficulty recognizing other people's autonomy
- intense control needs
- dissociation between idealized and devalued representations

Accordingly, developmental trauma should be understood broadly enough to include psychologically engulfing and autonomy-destroying environments, not merely overt physical violence.

9. The Same Developmental Environment Can Produce Different Trajectories

Neither trauma nor entitlement is sufficient by itself to determine the eventual manifold.

The developmental trajectory is better represented as:

Temperament + neurodevelopmental architecture + trauma + attachment + entitlement scripts + cultural scripts + reinforcement + social environment + individual choices

- emotional adaptation
- repeated behavior
- environmental feedback
- neural development
- increasingly stable manifold configuration

Thus even two children exposed to remarkably similar trauma can develop very different architectures.

One may develop:

State Specialization + Vigilance & Preparedness + Reflective Introspection

while another develops:

State Specialization + Agency & Directing + Antisocial

and another:

State Specialization + Systematizing + Vigilance & Preparedness

and another may develop a substantially different configuration altogether.

10. Boys and Girls Can Follow Either Trajectory

The gender distinction should therefore be understood probabilistically rather than categorically.

Boys

Because boys have historically received stronger culturally sanctioned dominance and entitlement scripts, developmental trauma may have a greater probability of interacting with those scripts to produce:

State Specialization + Agency & Directing

configurations.

This can occur even when the child was themselves scapegoated or violently abused.

Girls

Girls can absolutely develop the same configuration.

This may become more likely when they receive:

- familial entitlement
- golden-child treatment
- narcissistic projection
- engulfment
- enmeshment
- class-based entitlement
- racialized entitlement

- other culturally transmitted dominance scripts

Without sufficient entitlement reinforcement, however, developmental trauma may more often be organized toward:

State Specialization + Vigilance & Preparedness

or:

State Specialization + Reflective Introspection

or:

State Specialization + Avoidant/Social

configurations.

Again, these are probability shifts rather than deterministic rules.

The same principle applies to boys: sufficiently strong countervailing variables can produce any of these trajectories.

11. PTSD/C-PTSD Can Subsequently Modify the Underlying Manifold

Although PTSD/C-PTSD is not itself a manifold, sufficiently severe or prolonged trauma can change the trajectory of an underlying manifold.

This produces a recursive developmental relationship:

Underlying manifold → response to trauma → trauma adaptation → altered neural development → altered underlying architecture

Therefore trauma should not be treated as a superficial layer placed on top of an otherwise unchanged person.

Especially when trauma occurs during development, it can become constitutive of the eventual manifold configuration.

An adult may consequently possess a configuration that is neither simply:

"the original neurotype"

nor:

"PTSD added onto the original neurotype."

Instead, the trauma response may have become incorporated into the developmental architecture itself.

12. PTSD/C-PTSD as a Cross-Manifold Modifier

CMTR should therefore represent PTSD/C-PTSD as a cross-manifold modifier or developmental perturbation, rather than as an independent manifold.

Conceptually:

Underlying manifold(s)

↓

Traumatic exposure

↓

Trauma-response architecture

↓

Dissociation / hypervigilance / avoidance / intrusive memory / affective dysregulation / threat modeling

↓

Repeated adaptation

↓

Modification of underlying manifold trajectory

This permits PTSD/C-PTSD to occur alongside:

- Systematizing
- Vigilance & Preparedness
- Reflective Introspection
- State Specialization
- Interpersonal Prediction
- Agency & Directing
- Compulsive Engagement
- Salience Restructuring-related
- Epistemic Orientation configurations

without making PTSD itself an independent trait manifold.

13. Developmental Trauma and the State Specialization Manifold

The strongest general developmental prediction in this model is therefore:

Severe, repetitive, interpersonal, or inescapable childhood trauma increases the probability of a dominant or co-dominant State Specialization configuration.

The secondary manifold trajectory then depends substantially upon how the developing person learns to solve the problem of vulnerability.

Internalizing solution

"I must control myself."

Potential trajectory:

State Specialization + Vigilance & Preparedness

or

State Specialization + Reflective Introspection

or

State Specialization + Avoidant/Social

Externalizing/control solution

"I must control others."

Potential trajectory:

State Specialization + Agency & Directing

particularly where entitlement scripts, identification with the aggressor, reinforcement, and other developmental variables support that trajectory.

These are not mutually exclusive. An individual can move between strategies or contain both.

14. Emotional States and Neural Development

This model also reinforces a central CMTR principle:

Emotional states are not merely consequences of neural architecture; repeated emotional states participate in constructing that architecture.

The developmental feedback loop is:

emotional state

- behavioral adaptation
- environmental response
- reinforcement
- repeated neural activation
- plasticity and developmental change
- increased probability of the same future state

Thus:

chronic fear can strengthen threat-oriented processing;

chronic helplessness can strengthen dissociative and avoidance strategies;

chronic uncertainty can strengthen compulsive monitoring;

chronic entitlement and dominance reinforcement can strengthen Agency & Directing strategies;

chronic social rejection can strengthen withdrawal and rejection-sensitive processing.

The resulting adult architecture can therefore appear intrinsic even when it is substantially the product of developmental interaction.

15. Core CMTR Principle

PTSD and C-PTSD should therefore be represented as developmental consequences and modifiers that traverse the manifold system, not as independent manifolds.

The central developmental model is:

Predisposition + developmental environment → emotional state → adaptation → reinforcement → neural development → manifold trajectory.

And for developmental trauma specifically:

Developmental trauma → State Specialization organization is a common major pathway.

From there:

Trauma + strong entitlement scripts → greater probability of Agency & Directing organization.

while:

Trauma + insufficient entitlement scripts → greater probability of Vigilance & Preparedness, Reflective Introspection, or Avoidant/Social organization.

The final architecture is never determined by any one variable.

It emerges from the recursive interaction of biology, temperament, attachment, trauma, cultural narratives, entitlement scripts, reinforcement, social environment, and subsequent behavior.

This preserves one of CMTR's central principles:

The person is not the diagnosis, the trauma, or the manifold. The person is a dynamically developing trajectory through an interacting multidimensional field.

ADDITIONAL DEVELOPMENTAL TRAJECTORY — TRAUMA + JUSTICE → PROSOCIAL POWER/CONTROL

Developmental trauma can also produce a **prosocial Agency & Directing-dominant trajectory**.

A child who experiences severe injustice, coercion, helplessness, or abuse may develop an intense determination that:

No one should have to experience what I experienced.

Rather than responding to powerlessness by seeking domination over other people, the developing person may respond by developing a strong need to:

- protect vulnerable people
- prevent abuse
- establish fair systems
- expose wrongdoing
- enforce legitimate boundaries
- defend autonomy
- correct institutional failures

- create order where there was chaos
- acquire enough competence and authority to prevent recurrence of the original harm

The resulting configuration can therefore be:

State Specialization + Agency & Directing + Moral Intelligence

rather than:

State Specialization + Agency & Directing + Antisociality.

In this trajectory, the developmental adaptation is still fundamentally organized around **power and control**, but the object of control is different.

The antisocial trajectory tends toward:

“I must control other people so that I can never be vulnerable again.”

The prosocial trajectory tends toward:

“I must acquire enough power to prevent people from being harmed or systems from becoming abusive.”

The former contracts the autonomy of other agents.

The latter attempts to **protect or expand it**.

This is therefore a healthier incorporation of the trauma perturbation into the developing manifold.

It is not the absence of trauma effects. Rather, the trauma has been incorporated into a configuration in which the resulting Agency & Directing capacity is constrained by moral concern, autonomy recognition, and a desire to prevent recurrence.

DEVELOPMENTAL TRAUMA DOES NOT HAVE A SINGLE MANIFOLD OUTCOME

The major developmental pathways can therefore be represented as:

Trauma → Internalizing Adaptation

State Specialization + Vigilance & Preparedness

or

State Specialization + Reflective Introspection

or

State Specialization + Avoidant/Social

Core adaptation:

“I must control myself to become safe.”

Trauma → Antisocial Agency & Directing Adaptation

State Specialization + Antisocial Agency & Directing

Core adaptation:

“I must control others so that I can never be powerless again.”

Trauma → Prosocial Agency & Directing Adaptation

State Specialization + Prosocial Agency & Directing + Moral Intelligence

Core adaptation:

“I must acquire power and competence so that I can prevent people from being powerless or harmed.”

These trajectories can also overlap.

A person may simultaneously contain:

- substantial State Specialization organization
- Vigilance & Preparedness
- Reflective Introspection
- prosocial Agency & Directing
- antisocial Agency & Directing traits under particular conditions

The manifold model therefore describes **probabilistic developmental trajectories rather than discrete outcomes.**

THE NEAR-UNIVERSAL DEVELOPMENTAL EFFECT OF SEVERE CHILDHOOD TRAUMA

The strongest general prediction should therefore not be:

Developmental trauma produces antisociality.

Nor:

Developmental trauma produces anxiety/depression.

Nor:

Developmental trauma produces Agency & Directing.

Rather, the most general prediction is:

Severe, repetitive, interpersonal, or inescapable developmental trauma substantially increases the probability that State Specialization organization will become dominant or co-dominant.

The subsequent direction of manifold development is considerably more variable.

In simplified form:

Developmental trauma

↓

State Specialization organization

↓

secondary developmental trajectory

↓

Internalizing / Vigilance & Preparedness / Reflective Introspection

or

Antisocial Agency & Directing

or

Prosocial Agency & Directing

or other manifold configurations

The State Specialization response is therefore the relatively stable developmental consequence, while the **direction in which the rest of the manifold reorganizes is**

determined by the interaction of temperament, entitlement scripts, attachment, cultural narratives, reinforcement, moral development, social environment, and individual adaptation.

This distinction is important because it allows CMTR to explain why two people can experience comparably severe developmental trauma and both develop substantial dissociative organization while subsequently becoming radically different kinds of adults.

One may become:

fearful, avoidant, compulsive, and self-blaming.

Another:

dominating, entitled, and exploitative.

Another:

highly protective, justice-oriented, systems-focused, and determined to prevent others from experiencing similar harm.

All three can represent **different developmental solutions to the same fundamental problem of having learned that the world is unsafe.**

The difference lies in **where the developing system directs its adaptation: inward, outward toward domination, or outward toward protection and systemic correction.**

To link this to your first paper (RICD): have you considered using Fisher Information Metrics on this trait manifold? It would mathematically define the "distance" between cognitive profiles based on how easily a given environment can distinguish their behaviors. "Maladaptation" would then simply be regions of extreme informational curvature where feedback loops shatter.